

Building a Caterpillar Tunnel

Caterpillar tunnels are one of the simplest and most effective solutions for extending the market gardening season. In addition to extending the season, these tunnels can also be used to protect certain crops, such as field tomatoes, from wind and adverse weather conditions in summer.

The versatility of caterpillar tunnels not only increases production but offers a better return on your investment. Caterpillar tunnels are an especially affordable solution for the gardener who's just getting started.

Time Required

- 4 hours with 3 people

Tools

- Ground post driver
- Sledgehammer
- Staples
- Stepladder
- Adhesive tape
- Measuring tape
- Torch (if you need to cut geotextile)
- Ground cover anchoring pins
- Knife

Material

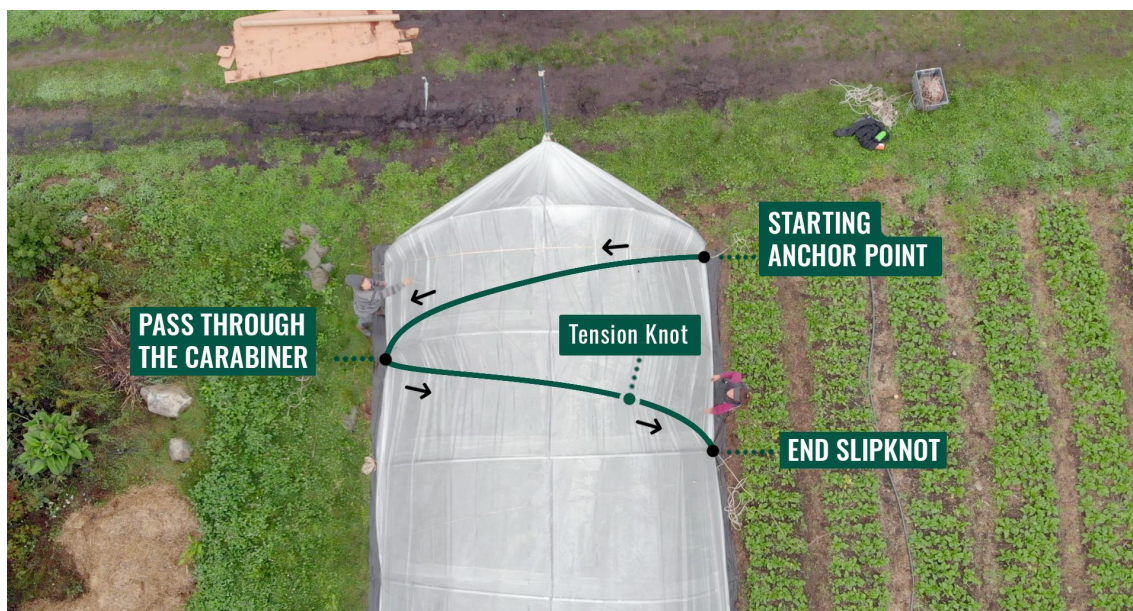
Caterpillar tunnel for 4 beds: 16 ft x 100 ft x 7.75 ft (5 m x 30 m x 2.4 m), hoops each 5 ft (1.5 m)	Nb of pieces	Price per piece \$CAD (\$USD)	Cost \$CAN (\$USD)
1 ¼ in (3.2 cm) galvanized steel hoops of 24 ft (7.3 m) long	20	\$69 (\$56)	\$1380 (\$1120)
1/2 po x 3.3 ft (15 mm x 1 m) concrete reinforcing bars (rebar)	40	\$3.25 (\$2.60)	\$130 (\$104)
1500 pi (460 m) of ¼ in UV-treated nylon or polyester rope	1	178\$ (\$148)	178\$ (\$148)
130 ft x at least 1 in (40 m long x at least 2.5 cm) wide polypropylene webbing	1	\$47 (\$38.70)	\$47 (\$38.70)
5/16 in (8 mm) < carabiners	40	\$1.10 (\$0.90)	\$44 (\$36)
Anchor plates with a 7/16 in (1 cm) hole for the carabiner and a 3/4 in (2 cm) hole for the rebar	40	\$5.16 (\$4.25)	\$206 (\$170)
6.5 ft (2 m) T posts	4	\$14 (\$11.50)	\$56 (\$46)
Webbing/strap tensioners	2	\$1.90 (\$1.60)	\$3.80 (\$3.20)
1.5 ft (0.45 m) x 100 ft (30 m) strips of woven ground cover	2	\$20.64 (\$17)	\$41.28 (\$34)
6 mm, UV-treated greenhouse plastic film 33 ft x 130 ft (10 m x 33,5 m) Width proportional to the hoops + 1 ft (30 cm) extra width on each side + 33 ft (10 m) in length compared to the beds	1	\$545 (\$450)	\$545 (\$450)
Ground cover anchoring pins	+/- 100	\$0.48 (\$0.40)	\$48 (\$40)
Total		\$CAD 2679.08 (\$USD 2189.90)	
Price per square foot (1600 ft² in total)	\$1.67 (\$1.37)		
Price per square meter (150 m² in total)	\$17.46 (\$14.60)		

Steps

- Staple the ground cover strips in the aisle where the hoops will be fixed.
- Make holes in the ground cover for each hoop, every 5 ft (1.5 m). Note that the first and last holes have to be 2.5 ft (0.8 m) inside the bed ends. It is best to use a blowtorch to keep the ground cover material from deteriorating quickly.
- Insert the anchor plates with their carabiners to the rebars.
- Drive in the rebars; leave 30 cm (12 in) sticking out of the soil.
- Slip the arches or hoops over the rebars. Start at the back of the bed and move forward.
- Drive in 2 T posts, 2 m (6,5 ft) from the first and last arches (4 T posts in total). Make sure they are next to each other and tilted at 60 degrees.
- With a little rope, attach the self-closing strap tensioners (strap clips) at the base of the T posts so they don't pull up.
- Insert the ropes in every second carabiners (or 10 on each side). The first time you will need to cut 20 pieces of rope 16.75 m (55 ft) long.
- Attach 2 ropes to each end hoop, at the $\frac{1}{3}$ and $\frac{2}{3}$ of the hoop. After, attach firmly the 2 ropes to the base of the T posts. Add tension with a tension knot (trucker's hitch). At the end, it will form a V shape.
- Pass the webbing through the strap tensioner. Using the stepladder, unroll the webbing by looping it around the apex of each arch, from the bottom up. Once at the end, pass the strap through the second tensioner to maintain tension.
- Then, beginning at the centre, tighten the webbing toward an end by pulling it down all along. At the end, tight it with the strap tensioner. Start again at the centre and do the same toward the other end. The installation of the strap can be done alone, but it's better with two

people. Both people start at the centre and pull the tension out toward each end of the caterpillar tunnel.

- If needed, install sprinkler irrigation along the arches, towards the inside of the structure. The irrigation line will run along the webbing.
- Slide the plastic from side to side. It takes at least 3 people to complete this task.
- 1 to 2 people per end will pull the plastic towards the T posts as much as they can to stretch it to the maximum. Pass the plastic between the T posts to catch it there, then twist the excess plastic on itself like a candy wrap.
- Keep pulling as much plastic as possible between the posts to ensure the end of the tunnel is secure and tight. Then, wrap the excess plastic around the base of the T posts and tie it firmly with rope. Close the top of the T posts with a knotted rope.
- Each piece of rope must be attached in a V shape (see picture below): Toss the rope from one side to the other, pull it through the next arch carabiner and send it back over to the third arch. The person who receives the rope tightens it using a tension knot (also called trucker's hitch), while the person on the other side makes sure the tension is equally distributed by pulling on it. The tensioning knot should be high enough to have some space for tightening the rope sufficiently to secure the tunnel in place. If the tensioning knot is too low, it will not be possible to stretch enough. Finish with a slipknot.





- Attach a new rope at the carabiner where the previous rope finishes and repeat the same operation until the end of the tunnel is reached. And do the same on the other side of the tunnel. In the end, the ropes should follow one another for the entire length of the tunnel. The ropes on either side of the tunnel should form Vs that intersect in reverse (as green and yellow lines in the picture above).
- Don't forget to re-tighten the ropes now and then, before storms or when high winds are forecasted.